

Continuous random variables

Recall,

Discrete random variable: Range of x = Countable setContinuous random variable

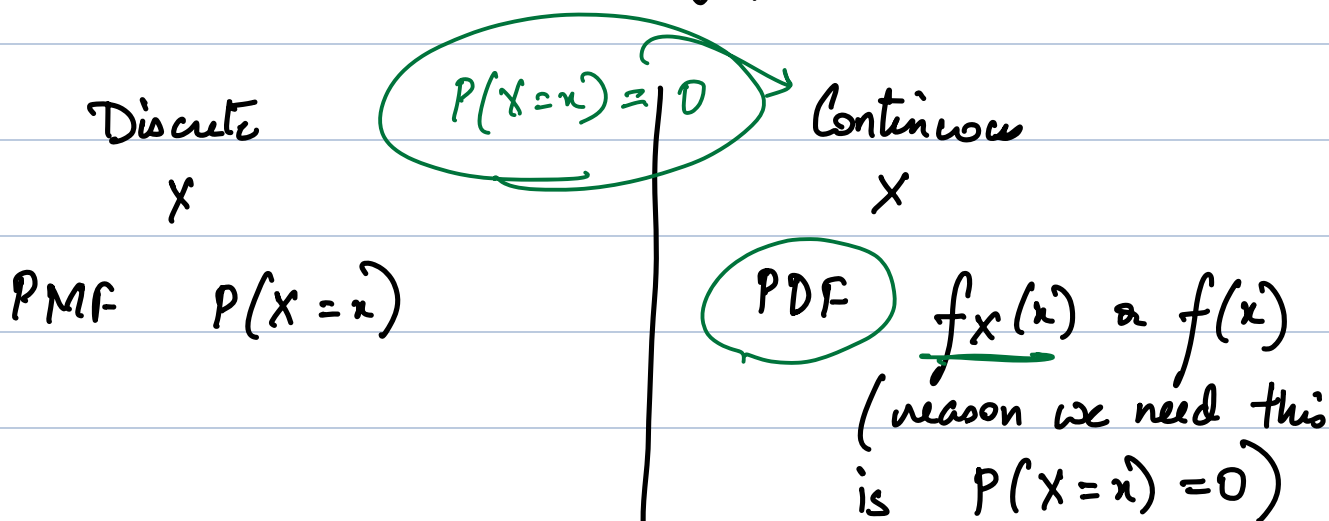
Random variables that take values from an uncountable set, e.g. real numbers or an interval

↳ or an infinite number of possible values within a given range or interval

also means "infinite outcome"

Eg. height of a person

time between arrivals of particles in a physics experiment



(A continuous r.v can take on an infinite number possible values within any range.

CDF $F_X(x) = P(X \leq x)$
(step funcⁿ)

Ex. Prove that CDF is continuous funcⁿ

CDF $F_X(x) = P(X \leq x)$

PDF (Probability density function)

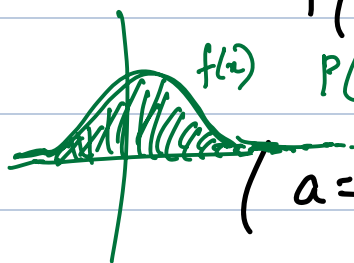
Not a probability!

$$P(a \leq X \leq b)$$

$$= \int_a^b f(x) dx$$

Defⁿ: A r.v. X has PDF $f(x)$ if

$$P(a \leq X \leq b) = \int_a^b f(x) dx \quad \text{for all } a \leq b$$



$$(a=b \Rightarrow \int_a^a f(x) dx = 0 \Rightarrow P(X=a)=0)$$

To be valid;

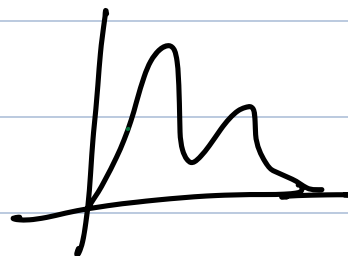
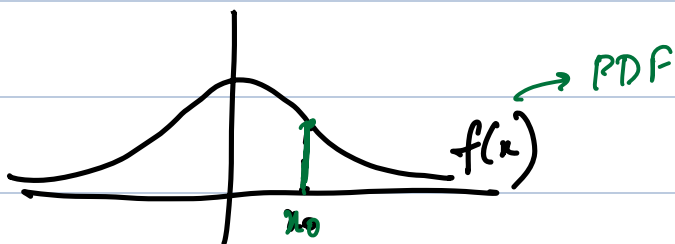
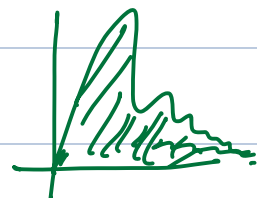
(i) $f(x) \geq 0$

(ii) $\int_{-\infty}^{\infty} f(x) dx = 1$

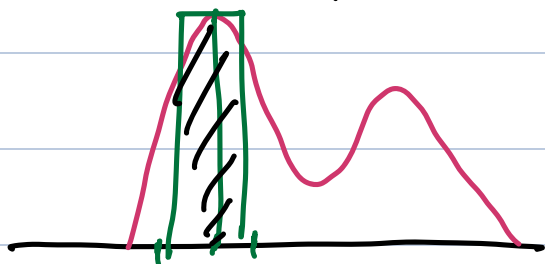
PMF

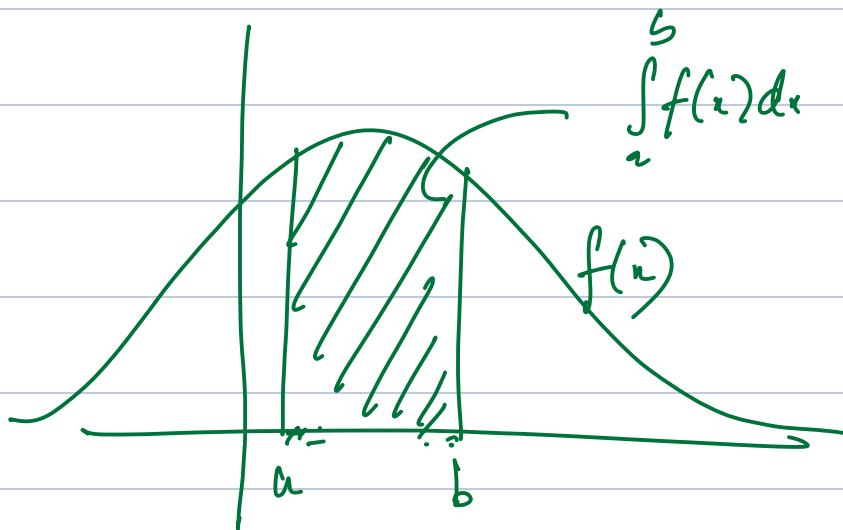
$$P(X=x) \geq 0$$

$$\sum_{R_j \in \mathcal{R}_X} P(X=x) = 1$$

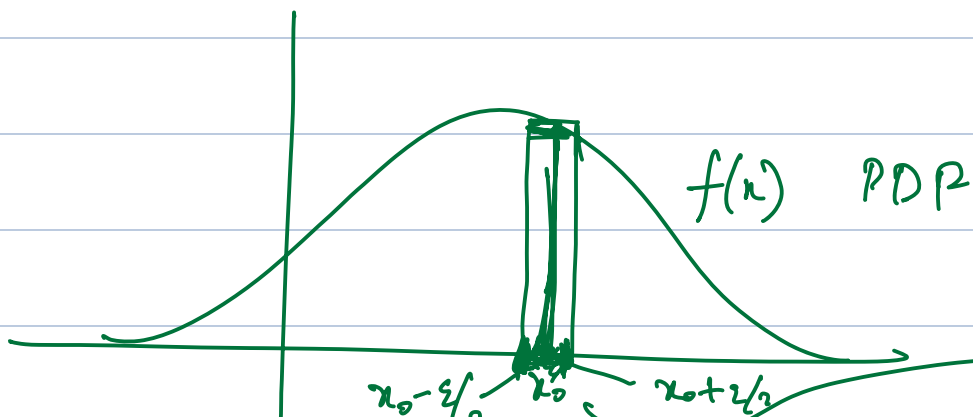


What does $f(x_0)$ mean?





$$P(a < X < b)$$



$$P\left(x_0 - \epsilon/2 < X < x_0 + \epsilon/2\right)$$

ϵ is very small

$$P(X = x_0)$$

$$P(X = x_0) \approx$$

$$= \epsilon \cdot f(x_0)$$

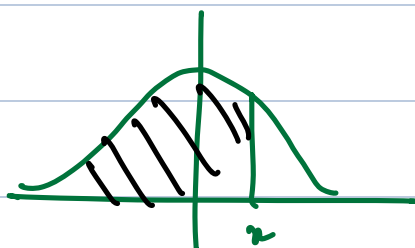
$$\begin{aligned} \text{area of the region} &= f(x_0) \cdot \varepsilon \approx \underbrace{P\left(x_0 - \frac{\varepsilon}{2} \leq X \leq x_0 + \frac{\varepsilon}{2}\right)} \\ &= P(X \approx x_0) \end{aligned}$$

$$P\left(X \in \left(x_0 - \frac{\varepsilon}{2}, x_0 + \frac{\varepsilon}{2}\right)\right)$$

$\varepsilon > 0$ very small

How is PDF related to the CDF? let X have PDF f

$$\text{CDF } F(x) \stackrel{\text{def.}}{=} P(X \leq x) = \int_{-\infty}^x \underline{f(t)} dt$$



$$P(-\infty < X \leq x) = \int_{-\infty}^x \underline{f(t)} dt$$

Other way around

Let X have CDF F (if X is continuous), Then

$$\text{PDF } f(x) = F'(x) = \int_a^b F'(x) dx \quad \text{FTC}$$

$$P(a < X < b) = \int_a^b f(x) dx = F(b) - F(a)$$

Discrete
 X

PMF $P(X=x)$

Continuous
 X

PDF $f_X(x)$ or $f(x)$

CDF $F_X(x) = P(X \leq x)$
(step funcⁿ)

$$E[X] = \sum x \underbrace{P(X=x)}$$

$$\text{Var}[X] = \underbrace{E[X^2]} - (E[X])^2$$

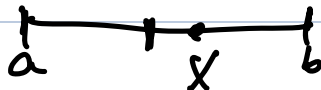
CDF $F_X(x) = P(X \leq x)$

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

Examples of continuous distribution

Uniform (a, b)



completely random point in $[a, b]$

Unif : probability $\propto$ length

PDF:

$$f(x) = \begin{cases} c, & \text{if } a < x < b \\ 0, & \text{otherwise} \end{cases}$$

$$P(a < X < b) = \int_a^b \underbrace{f(x)}_{\text{PDF}} dx$$

How to find 'c'

In this situation $\rightarrow \int_{-\infty}^{\infty} f(x) dx = 1$

$$\Rightarrow \int_a^b f(x) dx = 1 \Rightarrow$$

$$\int_a^b c dx = 1 \Rightarrow c = 1/(b-a)$$

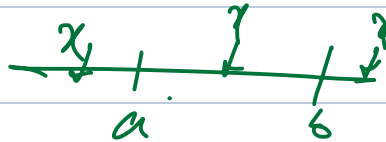
What is CDF

$F(x) = ?$

$$X \sim \text{Uniform}(a, b) \quad a \neq b$$

$$f_X(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{o/w.} \end{cases}$$

$$F(x) = \int_{-\infty}^x f_X(x) dx$$



$$F(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \leq x \leq b \\ 1 & x > b \end{cases}$$

What is $E[X]$? = $\frac{a+b}{2}$

$$\int_{-\infty}^{\infty} x f_X(x) dx$$

$$= \int_a^b x \cdot \frac{1}{b-a} dx$$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x) dx = \frac{a^2 + b^2 + ab}{3}$$

$$Var(X) = ? \quad \underbrace{(b-a)^2}_{12}$$

check!

Ex. Suppose the number of hours that a package gets delivered past noon is modelled by the following PDF:

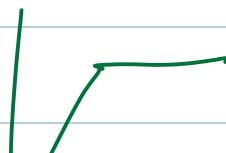
$$f_X(x) = \begin{cases} x/10 & , 0 \leq x < 2 \\ c & , 2 \leq x < 6 \\ 0 & \text{otherwise} \end{cases}$$

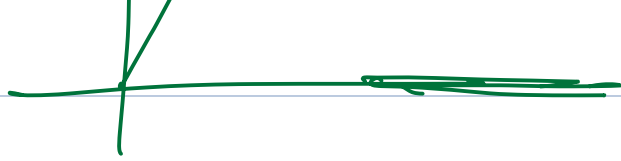
$P(X=x) = 0$

$[0, 6]$

Draw

① Graph of PDF





② What is Range of X ?

③ What is the value of 'c' that makes f_x a valid density function?

$$\int_{-\infty}^{\infty} f_x(x) dx = 1 \Rightarrow c = 1/5$$

④ Find CDF of X

$$F(x) = \int_{-\infty}^{\overset{(x)}{\quad}} \underline{\underline{f_x(x) dx}}$$

$$F_X(x) = \begin{cases} 0 & x < 0 \\ x^2/20 & 0 \leq x \leq 2 \\ x/5 - 1/5 & 2 < x \leq 6 \\ 1 & x > 6 \end{cases}$$

(5) What is the probability that the delivery arrives betⁿ 3 pm & 6 pm?

$$P(3 < X < 6) = \int_3^6 f(t) dt =$$

$$P(\text{---} < |a| < \text{---}) =$$

(6) What is the expected time that the package arrives at?

$$E[X] = \int_0^6 x f_X(x) dx$$

$$= \int_0^2 x \cdot \frac{x}{10} dx + \int_2^6 x \cdot \frac{1}{5} dx$$

Problem 2:

A 3-bit Analog-to-digital converter (ADC) operates over a full-scale voltage range of 0 V to 8 V.

The ADC maps continuous input voltages to the nearest discrete level.

The quantization error/noise (Q) is defined as the difference between the actual analog input & the quantized digital value.

Under specific conditions (busy signal), Q is modeled as a "Uniform Continuous random variable" on $[-\frac{\Delta}{2}, \frac{\Delta}{2}]$, where Δ : quantization step size (resolution).
Defining $\Delta = \frac{V}{2^N}$ (2^N)

- a) Calculate the step size Δ for this 3-bit system.
- b) " " PDF of noise
- c) What is the probability that the error is exactly 0.25V?
- d) "Noise Power" / Variance
- e) If the input voltage is restricted such that the error Q must stay within the range $[-0.2V, 0.2V]$, calculate
- $$P(|Q| > 0.15V) \quad \Bigg| \quad P(|Q| > 0.15 \mid |Q| \leq 0.2)$$

$$A = |Q| > 0.15, \quad B = |Q| \leq 0.2$$

Solution:

e) Find $P(|Q| > 0.15 \mid |Q| \leq 0.2)$

$$P(|Q| > 0.15 \mid |Q| \leq 0.2)$$

$$= \frac{P(|Q| > 0.15 \text{ \& } |Q| \leq 0.2)}{P(|Q| \leq 0.2)}$$

$$= \frac{P(0.15 < |Q| \leq 0.2)}{P(-0.2 \leq Q \leq 0.2)}$$

$$= \frac{P(-0.2 \leq Q \leq 0.2 \text{ \& } Q < -0.15 \text{ \& } Q > 0.15)}{P(-0.2 \leq Q \leq 0.2)}$$

$$= \frac{P(-0.2 \leq Q < -0.15) + P(0.15 < Q \leq 0.2)}{P(-0.2 \leq Q \leq 0.2)}$$

$$F(x) = P(X \leq x)$$

$$X \sim \text{Unif}(a, b)$$

$$\text{PDF: } f_X(x) = \begin{cases} \frac{1}{b-a} & \text{if } a < x < b \\ 0 & \text{otherwise} \end{cases}$$

$$P(a < x < b) = \int_a^b f(t) dt$$

$$\text{CDF: } F_X(x) = \begin{cases} 0 & \text{if } x < a \\ \frac{x-a}{b-a} & \text{if } a < x < b \\ 1 & \text{if } x \geq b \end{cases}$$

$$\text{Expectation: } E[X] = \frac{a+b}{2}$$

$$\text{Variance: } \text{Var}(X) = \frac{(b-a)^2}{12}$$

Uses of Uniform distribution.

Uniform is universal

Any continuous r.v. can be transformed into a uniform random variable, and conversely, a uniform random variable can be used to generate any continuous distribution.

$$F(x) = e^{-\lambda x}$$

$$F(X) = e^{-\lambda X}$$

Thm.: Let X be a continuous random variable with a CDF denoted as $F_X(x)$ or $F(x)$.

Let $Y = F_X(X)$ be a new random variable, then Y follows a Uniform distribution on $[0, 1]$.
 $Y \sim \text{Unif}(0, 1)$

Moreover, If U is a r.v. such that $U \sim \text{Unif}(0, 1)$ and F is a target distribution (any valid CDF function), then $X = F^{-1}(U)$ is a random variable with CDF of $X = F(x)$.

Proof: Let $U \sim \text{Unif}(0, 1)$ and $X = F^{-1}(U)$
Find CDF of X .
Show: CDF of $X = F(x)$

$$\begin{aligned} P(X \leq x) &= P(F^{-1}(U) \leq x) \\ &= P(U \leq F(x)) \end{aligned}$$

$$\text{If } U \sim \text{Unif}(0, 1)$$

$$\text{CDF of } U = P(U \leq y) = y$$

$$\Rightarrow P(X \leq x) = P(U \leq F(x)) = F(x)$$

$$\Rightarrow \text{CDF of } X = F(x)$$

Conversely, let $Y = F_X(X)$. To show: CDF of $Y =$

$$\begin{aligned} P(Y \leq y) &= P(F(X) \leq y) = P(X \leq F^{-1}(y)) \quad f(y) = y \\ &= F(F^{-1}(y)) = y \end{aligned}$$

$$\Rightarrow Y = F_X(X) \sim \text{Unif}(0,1)$$

Ex. Suppose we want to generate a r.v. X that follows an exponential distribution with rate λ .

$$\rightarrow F(x) = 1 - e^{-\lambda x}, x \geq 0$$

If a computer generates a uniform random number $U \sim \text{Unif}(0,1)$,
find X

Solⁿ: Target CDF: $F(x) = 1 - e^{-\lambda x}$ for $x \geq 0$
 X has CDF given by $F(x)$.

Given: U and $F(x)$

$$X = F^{-1}(U)$$

$$F(x) = 1 - e^{-\lambda x} \Rightarrow F^{-1}(x) = ?$$

$$t = 1 - e^{-\lambda x} \Rightarrow e^{-\lambda x} = 1 - t$$

$$\Rightarrow -\lambda x = \ln(1-t)$$

$$\Rightarrow x = -\frac{1}{\lambda} \ln(1-t)$$

$$F^{-1}(x) = -\frac{1}{\lambda} \ln(1-x)$$

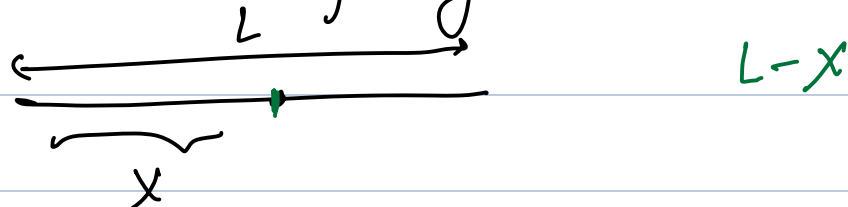
$$X = F^{-1}(U) = -\frac{1}{\lambda} \ln(1-U)$$

$U \rightarrow F$ has CDF $F(x) = 1 - e^{-\lambda x}$

$$(x > 0)$$

Ex. A stick is broken into 2 pieces, at a uniformly random breakpoint. Find the CDF and average of the length of the longer piece.

Sol.ⁿ: let $L =$ length of the stick



X : random breakpoint

$$Y = \max(X, L-X)$$

$$X \sim \text{Unif}(0, L)$$

$$Y \sim \text{Unif}\left(\frac{L}{2}, L\right)$$

Find $P(Y \leq y)$ ($F_Y(y)$)

$$= \begin{cases} 0 & y < \frac{L}{2} \\ \frac{y - \frac{L}{2}}{\frac{L}{2}} & \frac{L}{2} \leq y < L \\ 1 & y \geq L \end{cases}$$

PDF of Y : $f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{2}{L}$ for $\frac{L}{2} \leq y < L$

$$E[Y] = \int_{\frac{L}{2}}^L y \cdot f_Y(y) dy = \frac{3L}{4} \quad \text{Ans}$$

Normal distribution

Motivation : CLT

Def.~ Standard Normal distribution: A continuous r.v Z is said to have the standard Normal distribution if its PDF ϕ is given by

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad -\infty < z < \infty$$

$$Z \sim N(0, 1)$$